

PGRR: Planning-Guided Failure-Triggered Recovery and Rejoin for Dynamic Social Navigation

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Abstract—Classical navigation is reliable in routine mapped environments yet remains brittle under reciprocal pedestrian interactions. We introduce PGRR, a failure-triggered recovery layer that preserves Nav2 DWB for nominal navigation and invokes a learned policy only when observable risk or progress rules persist. Rather than commanding velocity, the policy selects among 21 temporary subgoals and four interpretable recovery modes. A planning mask rejects infeasible decisions, and an independent stopping-distance supervisor may override motion. A privileged short-horizon planner supplies recovery demonstrations for behavior cloning and DAgger; deployment uses only LiDAR and navigation history. Our locked Gazebo evaluation comprises 64 episodes across eight interaction families and three densities. On 24 matched Base-PGRR conditions, PGRR records no collision termination, compared with 79.2% for Base, a paired difference of -79.2 pp (95% bootstrap interval $[-91.7$ pp, -62.5 pp], Holm-adjusted $p_{\text{Holm}} < 0.001$). Goal success is 33.3% versus 20.8% ($p_{\text{Holm}} = 0.750$). On held-out recovery states, DAgger raises top-1 expert agreement from 88.5% to 92.2% and reduces expert-cost regret from 0.271 to 0.031. These results demonstrate an auditable route to contact-averse, interpretable recovery while identifying persistent reciprocal blockages as the principal remaining challenge.

I. INTRODUCTION

Classical navigation stacks remain attractive because mapping, global planning, local collision avoidance, and safety checks are explicit and independently testable. Their local planners can nevertheless freeze, oscillate, or enter reciprocal deadlock when moving pedestrians block a narrow passage. End-to-end learning can encode interaction patterns, but replacing the nominal controller also changes behavior in routine states where the classical stack is already competent.

This work asks whether learning can instead be confined to the short interval in which a classical planner is at risk of failure. PGRR (Planning-Guided Failure-Triggered Recovery and Rejoin) leaves Nav2 DWB in control during normal PointGoal navigation. An observable detector and a hysteretic state machine invoke recovery only for collision risk, freeze, oscillation, or deadlock. The policy never commands wheel velocity: it selects one of 21 local temporary goals or WAIT, BACKUP, REPLAN, and CONTINUE. Nav2 executes the decision, after which PGRR restores the original task goal. A planning-derived action mask and an independent braking supervisor constrain every learned decision.

Training exploits simulator state without leaking it to deployment. A short-horizon expert rolls out every legal recovery action using the map and privileged pedestrian state. Behavior cloning initializes a compact policy, and DAgger relabels states

visited by the learned controller. At test time, PGRR receives only LiDAR history, the local path and goal, robot and planner commands, progress history, planner status, and rule-detector scores.

The contributions are:

- a failure-triggered ROS2 hierarchy that preserves classical planning authority outside explicitly detected recovery intervals;
- a fixed, interpretable recovery action space with planning and sensor masks enforced during expert labeling and online policy execution;
- an automatically labeled BC-DAgger pipeline that aggregates learner-induced recovery states, with controlled ablations of aggregation and action masking; and
- a reproducible evaluation over eight pedestrian-interaction families, three densities, four recovery mechanisms, immutable raw streams, and paired statistical analysis.

Across 24 locked Base-PGRR pairs, PGRR changes the observed collision count from 19 to zero and the goal-completion count from five to eight. The evaluation separates contact avoidance, task completion, and intervention behavior so that these effects are not collapsed into a single score.

II. RELATED WORK

A. Classical and Social Navigation

The dynamic-window approach searches dynamically feasible velocity commands over a short horizon [1]; Nav2 provides a modern ROS2 navigation system in which DWB implements this family of local control [2]. Reciprocal methods such as ORCA [3] and learned crowd-aware policies such as SARL [4] instead model interaction more directly. The broader social-navigation literature emphasizes that collision avoidance, comfort, legibility, and task efficiency are distinct objectives [5]. PGRR retains DWB as the nominal controller and evaluates geometric clearance and discomfort time; it does not claim to model all social norms.

B. Hybrid Planning and Recovery

All-in-One learns a continuous switch among complete navigation planners [6], whereas DR-MPC learns a residual around model-predictive control [7]. Both demonstrate the value of hybrid model-based and learned navigation. PGRR differs in authority and output: intervention is failure-triggered,

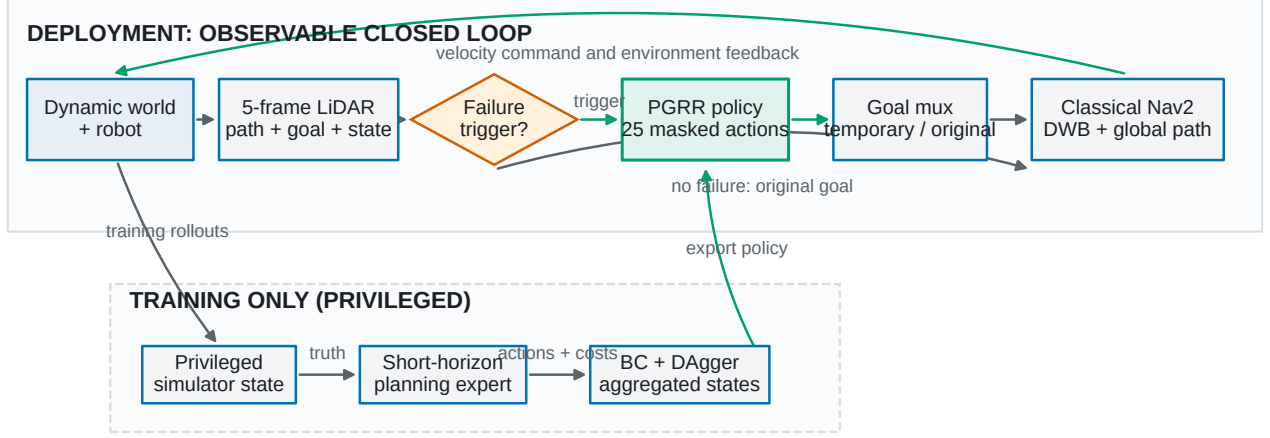


Fig. 1. PGRR architecture. Privileged simulator state and expert rollouts are confined to training (dashed region); deployment observes LiDAR and navigation history. Nav2 retains nominal and temporary-goal execution.

and the learned output is a temporary subgoal or an interpretable recovery mode rather than a nominal planner choice or velocity residual.

Navigation recovery itself is not new. Del Duchetto *et al.* learn local failure detection and recovery from human demonstrations [8]; Mohammad *et al.* predict planning failure and search for a recovery state using a Gaussian-process model [9]. Our focus is automatic privileged-planner labels, planning-mask enforcement, and aggregation on states induced by the recovery learner. DAgger supplies the standard reduction for this sequential distribution shift [10]. Invalid-action masking has also been analyzed in policy-gradient learning [11]; here it is used as a deterministic planning constraint for imitation inference, not as a safety proof.

C. Simulation and Evaluation

Arena-Bench introduced reproducible dynamic-obstacle evaluation [12], and later Arena releases expanded social-navigation scenarios and ROS2 support [13], [14], [15]. Our experiments use a pinned Humble/Gazebo backend and a deterministic pedestrian protocol; the later releases provide platform context rather than the experimental backend.

III. PROBLEM FORMULATION AND METHOD

A. Problem and Observable State

Let a differential-drive robot navigate a known static occupancy map from pose x_0 to goal g while a classical planner B supplies command u_t^B . PGRR must attempt recovery without using pedestrian truth at deployment. Its history o_t contains five 180-beam LiDAR scans, goal distance and bearing, eight path waypoints in the robot frame, robot velocity, u_t^B , ten-step goal-distance and angular-velocity histories, a one-hot planner status, and four rule scores. Ground-truth pedestrian state is isolated in a privileged record used only for expert labeling and evaluation.

B. Failure Trigger and Recovery State Machine

The detector computes collision-risk, freeze, oscillation, and deadlock scores. Collision risk combines braking distance, 1.5-s time to collision, absolute near-field clearance, and a 0.5-s radial closing trend. Freeze requires more than 1 m remaining, less than 0.15 m displacement over 3 s, and a planner request for motion. Oscillation requires at least six angular-velocity sign changes after a 0.08 rad/s deadband and less than 0.20 m progress over 4 s. Deadlock uses less than 0.15 m displacement and less than 0.15 m goal progress over 4 s, an obstacle within 1.2 m, and low robot speed. Motion rules are disabled for the first 4 s after the synchronized episode handshake; collision evidence remains active.

The usual state sequence is NORMAL→PENDING→RECOVERY→REJOIN, although a confirmed one-frame trigger may enter RECOVERY directly after cooldown. EMERGENCY-STOP has highest priority and may resume; only FAILED and SUCCEEDED are terminal. Recovery enters above $\tau_{on} = 0.65$ and exits after four frames below $\tau_{off} = 0.35$ plus valid goal progress. Decisions occur at 2 Hz, velocity control at 10 Hz, with a 0.5-s minimum hold, 2-s cooldown, an 8-s learned-policy recovery bound, a 5-s rejoin bound, and four consecutive recovery attempts. The original goal is stored before intervention and reissued whenever a bounded temporary option hands control to REJOIN; this transition does not require Nav2 to have reached the temporary target.

C. Recovery Actions, Mask, and Safety Supervisor

The 25 actions are 21 robot-relative subgoals

$$(r, \theta) \in \{0.6, 1.0, 1.4\} \text{ m} \times \{-90, -60, -30, 0, 30, 60, 90\}^\circ, \quad (1)$$

and WAIT, BACKUP, REPLAN, and CONTINUE. For action a , a Boolean mask $M(o_t, a)$ rejects occupied or off-map endpoints, disconnected grid components, an obstructed swept Li-

DAR segment, insufficient rear observability, excessive global-path deviation, or unavailable replanning. WAIT is the final safe fallback, so at least one action remains legal. Before softmax, masked logits are set to the minimum representable value.

An independent supervisor may override any policy output. Its longitudinal stopping distance is

$$d_{stop} = \frac{v^2}{2a_b} + v\ell + m, \quad (2)$$

with $a_b = 0.8 \text{ m/s}^2$, latency $\ell = 0.15 \text{ s}$, and margin $m = 0.45 \text{ m}$ in the recovery controller. It provides bounded stop, reverse, turn, and forward escape submodes, all logged separately from the learned action. This is an empirical safety filter, not a formal guarantee.

D. Privileged Planning Expert

At training time the expert constructs local occupancy from the recorded LiDAR and observes privileged robot and pedestrian positions. Pedestrian velocities are estimated by finite differences rather than read as simulator truth. It uses constant-velocity prediction with conservative yielding and uncertainty inflation, then simulates every legal action for 3 s at 0.1-s resolution. Subgoal rollouts use grid A* and pure pursuit with differential-drive kinematics; the four mode actions have matched short-horizon models. The cost is

$$J(a) = w_c J_c + w_p J_p + w_s J_s + w_r J_r + w_l J_l \quad (3)$$

$$+ w_{sm} J_{sm} + w_t J_t + w_{sw} J_{sw} + w_w J_w, \quad (4)$$

where the terms denote collision, insufficient progress, social-distance integral, path-rejoin distance, length, angular smoothness, time, side switching, and repeated waiting. The weights are respectively $(10^6, 8, 4, 3, 0.4, 0.2, 0.3, 0.5, 1)$; collision and masked actions are effectively hard constraints. Besides the minimum-cost action, the expert stores all 25 costs and the gap to the second-best action for analysis.

E. Behavior Cloning and DAgger

The policy uses two 1-D convolutions over the 5×180 LiDAR stack, a 128-unit state encoder, and a two-layer head, totaling 59,193 parameters. For the selected model, uniform behavior cloning minimizes

$$\mathcal{L}_{BC} = -\frac{1}{N} \sum_i \log \pi_\theta(a_i^* | o_i, M_i). \quad (5)$$

DAgger then executes the learner on training scenarios, asks the privileged expert to relabel visited recovery states, aggregates them, and retrains. The selected aggregate contains 1,387 raw recovery states from 12 episodes; mirror augmentation yields 2,774 training samples, with 399 states in a scenario-disjoint validation set. Training uses AdamW at 3×10^{-4} , batch size 128, maximum 30 epochs, gradient norm 1, and early-stopping patience 6. The deployment checkpoint is selected by validation loss. Its configuration uses uniform BC and no PPO; throughout the paper, PGRR therefore denotes the selected DAgger policy rather than an RL-refined variant.

IV. EXPERIMENTAL SETUP

A. Platform and Robot

Experiments use a pinned ROS2 Humble Arena Docker image (Arena commit c2ff4a87) with the Gazebo backend, Nav2 NavFn/DWB, and a Jackal model. Scenario-coordinate bounds span $31.28 \times 24.03 \text{ m}$. The evaluation footprint radius is 0.36 m; the loaded DWB limits are 0.26 m/s and 1.0 rad/s. A 360-degree, 360-sample planar GPU LiDAR runs at 10 Hz and is resampled to 180 bins. Pose-derived simulation odometry supplies the declared known-pose interface, while evaluation independently uses Gazebo model poses for physical goal and contact checks.

To make pedestrian motion exactly replayable and LiDAR-observable, the protocol uses contactless cylindrical actors following deterministic scenario routes. Most actors hold when a candidate step would both enter a 1.3 m robot-clearance region and reduce separation; temporary blockage uses a finite route and a 0.8 m threshold. This construction avoids rigid-body impulses while preserving dynamic occlusion and reciprocal passage conflicts.

B. Scenarios, Splits, and Locked Manifest

Eight families exercise distinct local failures: head-on corridor, doorway bottleneck, crossing flow, blind corner, group blocking, overtaking, opposite streams, and temporary blockage. Low, medium, and high densities contain 2, 4, and 6 pedestrians, with family-specific speed ranges of 0.08–0.75 m/s. Train, validation, and test use disjoint compiled scenario variants and seed ranges 1000, 2000, and 3000, respectively.

Before test execution we tagged commit 35d7e60 as pre-final-eval, hashed the scenario manifest, checkpoint, detector, state-machine, and action configurations, and prohibited further tuning. The primary paired test runs Base DWB and PGRR on all eight families and three densities (48 episodes). At high density, Standard Nav2 Recovery and the rule-triggered Heuristic hierarchy add 16 episodes, for 64 algorithm episodes total. Each family-density cell has one fixed test seed, shared by all methods included in its paired comparison.

C. Baselines and Ablations

The evaluated closed-loop methods are: (i) Base DWB with periodic global replanning and no recovery subtree; (ii) Arena’s Standard Nav2 recovery behavior tree; (iii) the same failure trigger, state machine, action mask, and safety supervisor with deterministic Heuristic action selection; and (iv) PGRR with the selected DAgger policy. The primary pair isolates addition of the complete hierarchy to Base. The high-density four-method subset separates standard, heuristic, and learned recovery mechanisms.

The training workflow completes two aggregation rounds on the train split. Validation selects the first-round aggregate augmented by one train-only head-on coverage shard. The selected aggregate contains 1,387 recovery states from 12 episodes, and no test episode enters model selection.

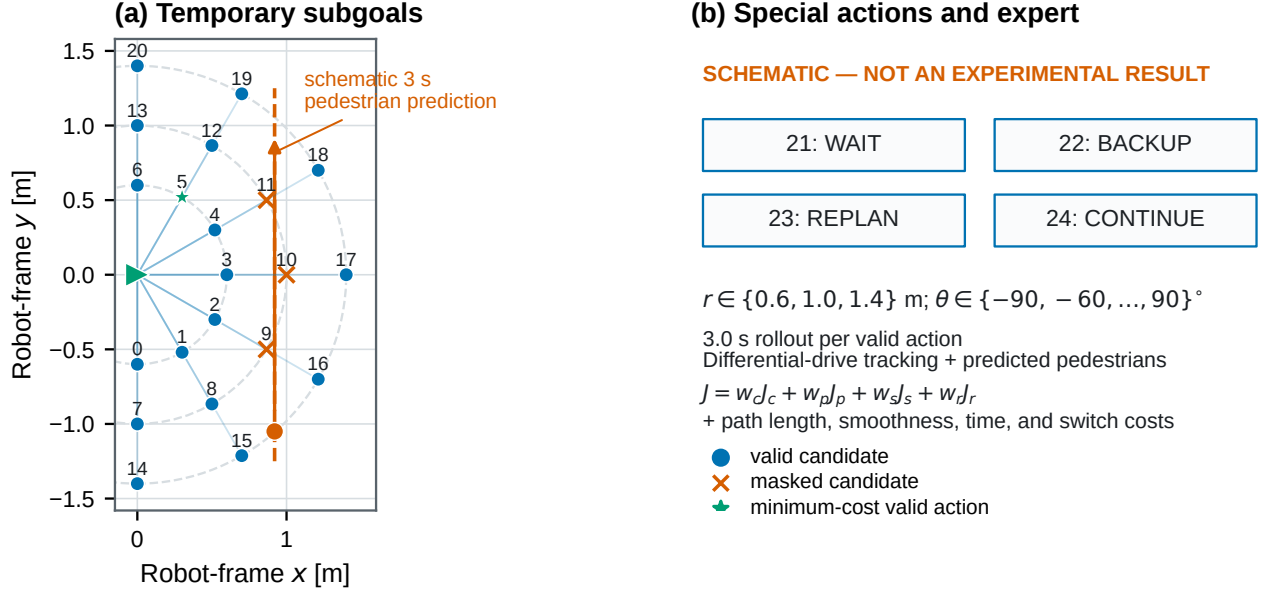


Fig. 2. Interpretable recovery decisions and privileged expert labeling. Candidate rollouts that violate the configured planning mask are never selectable; privileged pedestrian prediction is used only to rank legal training actions.

A separate offline ablation evaluates Uniform BC, Margin-Weighted BC, and the selected DAgger checkpoint on the same 399 scenario-disjoint recovery states. Each model is evaluated both with the original mask and with an all-legal mask; the latter reports unsafe proposals only and is never executed in simulation.

D. Metrics and Statistical Protocol

Every 10-Hz stream records robot and pedestrian geometry, commands, planner and recovery state, failure scores, LiDAR, and paths. We report terminal success, collision, timeout, planner failure, SPL, physical path length, navigation time, minimum human-centre distance, time below 1.0 m, fraction below 1.2 m, angular jerk, emergency stops, triggers, recovery duration, and intervention ratio. SIMULATOR_FAILURE and INVALID_RESET attempts are retained, counted, and retried once, but excluded from algorithm metrics. All other failures remain.

Base and PGRR are paired by the same immutable scenario and seed. Binary outcomes use exact McNemar tests; continuous paired differences use two-sided Wilcoxon signed-rank tests. Mean differences receive 10,000-sample paired bootstrap 95% confidence intervals, and related p -values use Holm correction. We report effects and confidence intervals even when significance is absent.

E. Execution Integrity

Concurrent workers use distinct ROS domain IDs and Gazebo partitions. A start handshake holds pedestrians until navigation and logging are active. Goal reach requires a physical-pose confirmation; supported shelves use a robot-circle-to-oriented-box distance test. Near-pedestrian LiDAR

consistency is evaluation-only. The runner refuses overwrite, preserves both physical attempts after an infrastructure retry, and fails unless each logical manifest row has exactly one valid algorithm outcome. Tables and figures are regenerated from the resulting Parquet/JSON artifacts rather than hand-entered values.

V. RESULTS AND ANALYSIS

A. Closed-Loop Navigation Outcomes

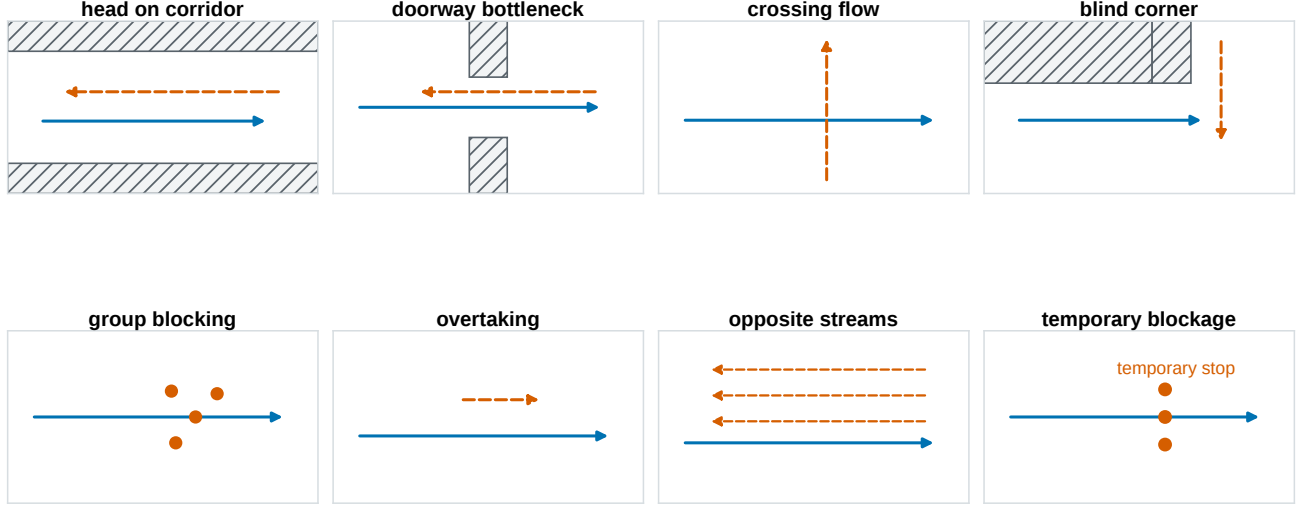
Table I reports every retained algorithm outcome. Across the 24 matched family-density conditions, Base DWB reaches 20.8% of goals and collides in 79.2%; PGRR reaches 33.3% and records no collision termination. Its mean SPL is 0.303 versus 0.205 for Base. Figure 4 retains the density and family structure that a pooled rate can hide.

The dominant paired effect is the collision reduction, -79.2 pp [-91.7 pp, -62.5 pp] ($p_{\text{Holm}} < 0.001$). The success difference is $+12.5$ pp [$+0.0$ pp, $+29.2$ pp] and is not statistically resolved ($p_{\text{Holm}} = 0.750$). Terminal non-success shifts from contact to horizon termination: the timeout difference is $+66.7$ pp [$+45.8$ pp, $+83.3$ pp] ($p_{\text{Holm}} < 0.001$). Thus, this operating point provides a statistically supported collision reduction, but not a supported improvement in completion. Table II reports all paired effects and continuous tests with multiplicity correction.

B. Safety, Efficiency, and Intervention

Mean minimum human-centre distance is 1.318 m for Base and 1.267 m for PGRR. Their paired difference is -0.051 m [-0.498 m, $+0.251$ m] with $p_{\text{Holm}} = 0.257$. Among episodes where both methods reach the goal, the mean PGRR-minus-Base navigation-time difference is $+9.4$ s [-2.1 s, $+30.8$ s]

Eight deterministic social-navigation stress families



Solid blue: robot route; dashed orange/circles: pedestrians; hatched gray: walls. Three densities; main paired set $n=24$ episodes/method; schematics not to scale.

Fig. 3. The eight locked test families. Each panel is generated from its checked-in scenario definition and shows the static layout, robot start and goal, reference route, and initial pedestrian motion.

TABLE I
CLOSED-LOOP NAVIGATION OUTCOMES ON THE LOCKED MANIFEST.

Method	n	Success $\uparrow$ (%)	Collision $\downarrow$ (%)	Timeout $\downarrow$ (%)	SPL $\uparrow$	Time $\downarrow$ (s)	$d_{\min}$ $\uparrow$ (m)	Excl.
DWB	24	20.8	79.2	0.0	0.205 ± 0.409	85.6 ± 10.7	1.32 ± 1.31	1
Standard	8	12.5	87.5	0.0	0.106 ± 0.299	114.5	1.01 ± 0.95	0
Heuristic	8	12.5	0.0	87.5	0.108 ± 0.306	129.4	1.09 ± 0.31	0
PGRR	24	33.3	0.0	66.7	0.303 ± 0.440	111.9 ± 32.2	1.27 ± 0.58	2

Note. DWB and PGRR use the same $n = 24$ family-density episodes; Standard and Heuristic use the common high-density subset ($n = 8$). Rates are computed over valid episodes. SPL and $d_{\min}$ use all valid episodes; Time uses successful episodes only. Continuous entries are mean $\pm$ sample SD when at least two observations exist. Excl. counts retained simulator/reset failures that were retried and excluded from algorithm rates. Bold identifies the proposed method.

($p_{\text{Holm}} = 1.000$). The raw successful-run means are 85.6 s and 111.9 s. Figure 5 compares all four methods on their common high-density subset, while Table III shows the resulting clearance-time trade-off, emergency interventions, recovery triggers, and rejoin success. Neither minimum human distance nor discomfort time differs after Holm correction. PGRR invokes more emergency stops and spends more time under recovery control, consistent with the hierarchy being active rather than merely reproducing nominal DWB.

C. Recovery Dynamics and Run Evidence

Figure 6 aligns failure score, goal progress, recovery state, and selected actions for a representative final episode chosen by a deterministic artifact rule. Figure 7 reconstructs three instants from that episode’s 10-Hz telemetry: robot and pedestrian poses, the global path, and recovery state. It is explicitly a telemetry reconstruction rather than a simulator camera image. Together they verify that the policy intervenes transiently and

that the original goal is restored; they are mechanism evidence, not additional trials.

D. Imitation and Mask Ablations

Table IV compares Uniform BC, Margin-Weighted BC, and DAGger on the same 399 scenario-disjoint validation states. In this dataset, margin weighting does not change the reported decision metrics and is therefore not part of the selected method. DAGger reduces expert-cost regret and improves top- k agreement. Disabling the mask produces many invalid proposals, so the no-mask rows are an offline diagnostic and were never executed on the robot. These results support distribution aggregation and mask enforcement, but they do not isolate the online safety supervisor from the learned policy.

E. Interaction-Condition Analysis

PGRR completes every crossing-flow and group-blocking condition and two of three temporary-blockage conditions. The remaining horizon-limited runs cluster in head-on, doorway,

TABLE II
PAIRED PGRR–DWB COMPARISONS ON THE LOCKED MANIFEST.

Metric	Test	Δ [95% CI]	p_{raw}	p_{Holm}	Effect size	n_{pairs}
Goal reached	McNemar	+12.5 [+0.0, +29.2] pp	0.250	0.750	$\text{OR}_H = 7$	24
Collision	McNemar	−79.2 [−91.7, −62.5] pp	< 0.001	< 0.001	$\text{OR}_H = 0.0256$	24
Timeout	McNemar	+66.7 [+45.8, +83.3] pp	< 0.001	< 0.001	$\text{OR}_H = 33$	24
Success time	Wilcoxon	+9.4 [−2.1, +30.8] s	1.000	1.000	$r_{\text{tb}} = +0.067$	5
Success path	Wilcoxon	+0.63 [−0.44, +2.36] m	1.000	1.000	$r_{\text{tb}} = −0.067$	5
SPL	Wilcoxon	+0.098 [−0.004, +0.222]	0.173	0.692	$r_{\text{tb}} = +0.619$	24
Min. human distance	Wilcoxon	−0.051 [−0.498, +0.251] m	0.037	0.257	$r_{\text{tb}} = +0.487$	24
Personal-space violation	Wilcoxon	+5.1 [+0.2, +10.2] pp	0.058	0.349	$r_{\text{tb}} = +0.509$	24
Discomfort time	Wilcoxon	−0.69 [−2.36, +1.13] s	0.135	0.677	$r_{\text{tb}} = −0.412$	24
Emergency stops	Wilcoxon	+14.33 [+10.79, +17.79] ep. ^{−1}	< 0.001	< 0.001	$r_{\text{tb}} = +1.000$	24
Recovery triggers	Wilcoxon	+5.21 [+3.92, +6.58] ep. ^{−1}	< 0.001	< 0.001	$r_{\text{tb}} = +1.000$	24
Intervention ratio	Wilcoxon	+58.0 [+46.2, +68.8] pp	< 0.001	< 0.001	$r_{\text{tb}} = +0.987$	24
Angular jerk	Wilcoxon	+0.575 [+0.407, +0.748] rad s ^{−3}	< 0.001	< 0.001	$r_{\text{tb}} = +0.980$	24

Note. All differences are PGRR minus DWB and use 95% percentile-bootstrap intervals; rate differences are in percentage points (pp). Binary endpoints use exact McNemar tests and the Haldane-corrected matched odds ratio OR_H ; continuous endpoints use Wilcoxon signed-rank tests and matched rank-biserial correlation r_{tb} . Holm correction covers all endpoints in this table. The paired sample size varies for success-conditional metrics. Comparison set: DWB vs PGRR.

TABLE III
RECOVERY BEHAVIOR ON VALID EPISODES FROM THE LOCKED MANIFEST.

Method	n	Triggers (ep. ^{−1})	Recovery success (%)	Recovery time (s/ep.)	Intervention (%)	Emergency stops (ep. ^{−1})
DWB	24	0.0 ± 0.0	–	0.0 ± 0.0	4.0 ± 3.9	0.0 ± 0.0
Standard	8	0.0 ± 0.0	–	0.0 ± 0.0	5.0 ± 4.0	0.0 ± 0.0
Heuristic	8	6.1 ± 4.1	62.9 ± 31.6	132.5 ± 40.6	75.7 ± 19.6	14.1 ± 8.1
PGRR	24	5.2 ± 3.5	73.6 ± 15.3	105.3 ± 58.4	61.9 ± 30.0	14.3 ± 8.9

Note. Entries are episode-level mean ± sample SD. Recovery time is the cumulative time spent in recovery per episode. Undefined success rates for methods with no recovery trigger are shown as dashes. Bold identifies the proposed method.

TABLE IV
OFFLINE POLICY ABLATION ON 399 SCENARIO-DISJOINT VALIDATION STATES.

Model	Planning mask	Top-1 (%)	Top-3 (%)	Invalid (%)	Cost regret	Near-opt. (%)	Catastrophic (%)
Uniform BC	Enabled	88.5	95.7	0.0	0.271	88.7	0.0
Uniform BC	Disabled [†]	0.0	91.0	93.2	9.32×10^5	0.0	93.2
Margin-weighted BC	Enabled	88.5	95.7	0.0	0.271	88.7	0.0
Margin-weighted BC	Disabled [†]	0.0	91.7	93.2	9.32×10^5	0.0	93.2
Triggered Dagger	Enabled	92.2	98.5	0.0	0.031	93.2	0.0
Triggered Dagger	Disabled [†]	12.0	41.9	84.0	8.40×10^5	12.8	84.0

Note. Regret is measured in expert-cost units; its hard invalid-action penalty dominates mask-disabled rows. [†] Disabled-mask rows are counterfactual offline proposals and were never executed in closed loop. “Catastrophic” denotes selection of a hard-penalty action. Bold identifies the selected policy.

blind-corner, opposite-stream, and overtaking interactions. The generated failure taxonomy separates nine terminally stationary runs from seven runs that retain short-window progress, including every blind-corner case. This split points to two distinct extensions—deadlock resolution and horizon-aware progress reasoning—without attributing either behavior to a specific action unavailable in the aggregate result table.

VI. LIMITATIONS

The study has several important limits. First, the sensing interface is planar LiDAR on a known two-dimensional map, and simulation localization is more accurate than a deployed AMCL stack. Second, the Gazebo fallback pedestrians follow deterministic routes and use a simple yielding rule; they do not

capture human intent, group norms, perception uncertainty, or the diversity of real crowds. Minimum distance and discomfort time are therefore geometric proxy metrics, not validated human-comfort judgments.

Third, the current operating point prioritizes contact suppression: 16 of 24 PGRR primary trials terminate at the evaluation horizon. Because the detector, mask, policy, and supervisor act jointly, this study cannot assign that behavior to one component. The privileged expert also depends on simulator truth and a 3-s constant-velocity rollout, which cannot represent prolonged reciprocal negotiation; the finite action set and safety supervisor provide no formal collision-avoidance guarantee.

The offline labeler and Heuristic baseline retain a legacy

(a) Goal completion by family (three densities)

head on corridor	0/3	0/3
doorway bottleneck	0/3	0/3
crossing flow	3/3	3/3
blind corner	0/3	0/3
group blocking	0/3	3/3
overtaking	0/3	0/3
opposite streams	0/3	0/3
temporary blockage	2/3	2/3
	DWB	PGRR

(b) Completion and collision endpoints by density

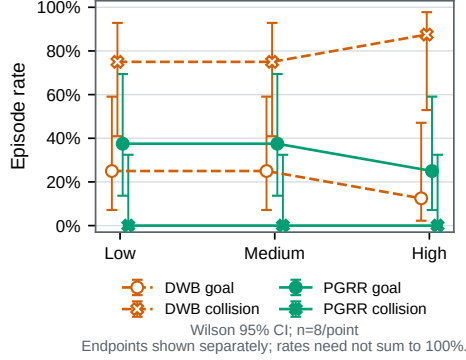


Fig. 4. Goal completion and collision endpoints. Panel (a) reports goals reached out of three densities for each family. Panel (b) reports each endpoint separately with Wilson 95% intervals ($n = 8$ per point); the two rates need not sum to one because horizon terminations remain possible.

270-degree angular calibration for 180 bins sampled from the 360-degree sensor. The deployed mask uses scan meta-data, and the CNN receives identical full-scan resampling during training and test, but expert action costs may inherit this distortion. We preserve the frozen evaluation instead of altering the pipeline after test-set inspection. Finally, each family–density cell has one fixed seed. The 24 primary pairs support cross-condition paired inference, not a low-variance per-scenario estimate. Real-robot validation, responsive human models, learned failure prediction, and PPO refinement remain outside the evaluated scope.

VII. CONCLUSION

We presented PGRR, a recovery interface that retains DWB for nominal control and restricts learned intervention to interpretable temporary goals and recovery modes. A privileged rollout planner supplies supervision, DAGger covers learner-induced states, and planning masks constrain both labeling and online inference. Across 24 locked Base–PGRR pairs, observed collision terminations decrease from 19 to zero while goal completions increase from five to eight; on held-out states, DAGger also improves expert agreement and cost regret. These results establish a reproducible foundation for failure-triggered recovery. Longer-horizon interaction reasoning, multi-seed

High-density safety–efficiency (paired episodes)

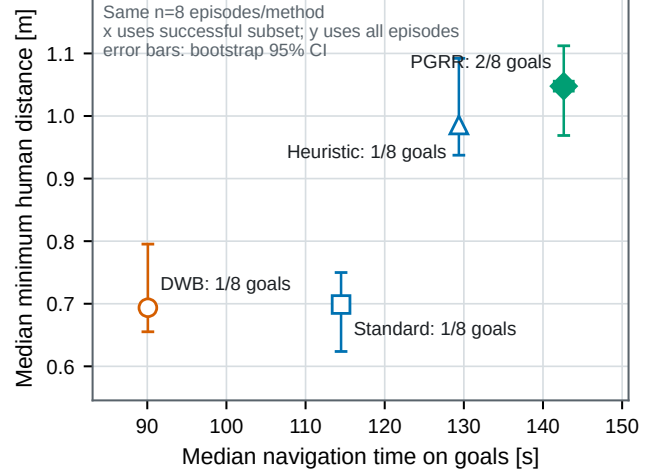


Fig. 5. Safety–efficiency view on the common eight-condition high-density subset. Marker shape encodes terminal outcome, keeping completion status and episode duration visible alongside geometric clearance.

Measured PGRR timeline — group_blocking / high (n=1)

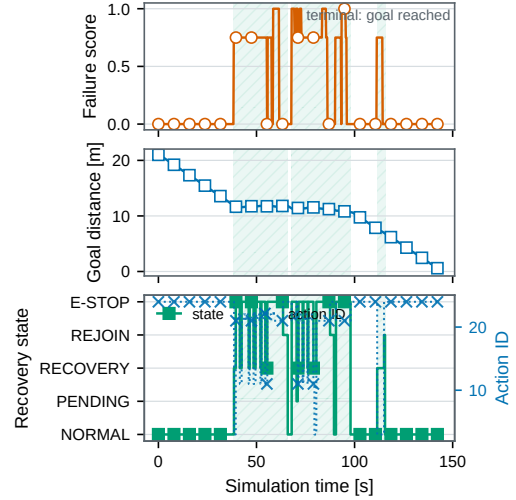


Fig. 6. Recorded failure score, goal distance, and recovery state for a representative final episode. All event times come from the raw JSONL stream.

replication, richer pedestrian models, and hardware validation are the most direct next steps.

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Logged PGRR runtime sequence — group_blocking / high (successful triggered recovery)

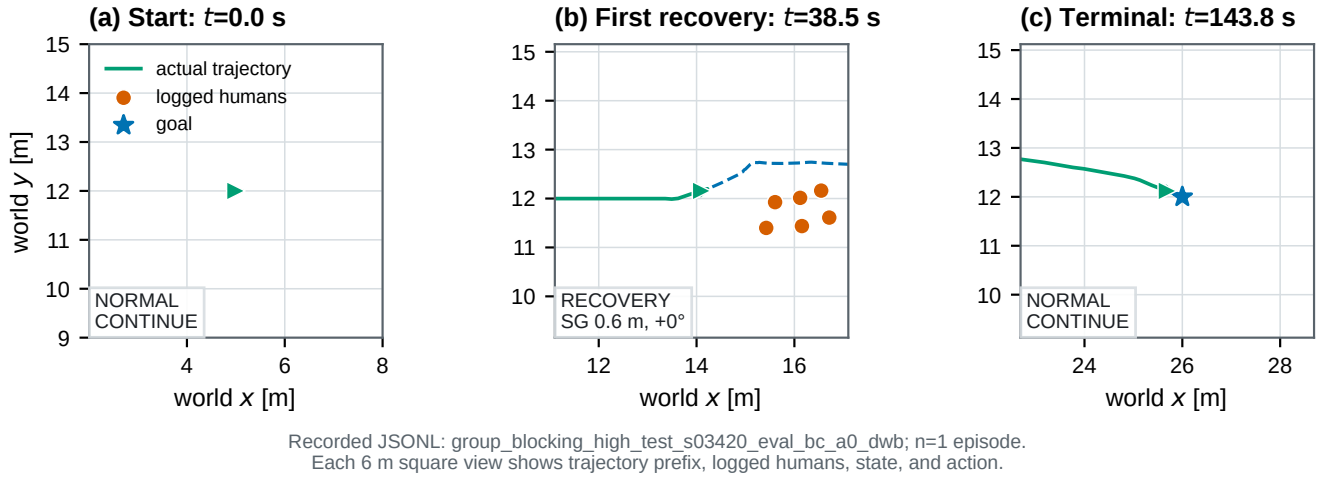


Fig. 7. Telemetry reconstruction at episode start, first recovery intervention, and termination. Circles are recorded physical pedestrian centres; the robot pose, path, action, and state are taken from the same raw stream.

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